

Multipocket Cage Enables the Binding of High-Order Bulky and Drug Guests Uncovered by MS Methodology

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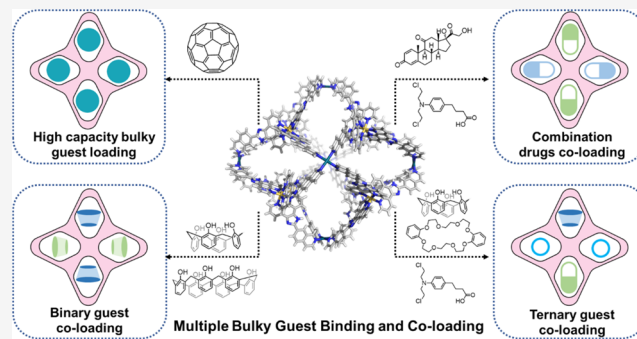
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ABSTRACT: Achieving high guest loading and multiguest-binding capacity holds crucial significance for advancement in separation, catalysis, and drug delivery with synthetic receptors; however, it remains a challenging bottleneck in characterization of high-stoichiometry guest-binding events. Herein, we describe a large-sized coordination cage (MOC-70-Zn₈Pd₆) possessing 12 peripheral pockets capable of accommodating multiple guests and a high-resolution electrospray ionization mass spectrometry (HR-ESI-MS)-based method to understand the solution host–guest chemistry. A diverse range of bulky guests, varying from drug molecules to rigid fullerenes as well as flexible host molecules of crown ethers and calixarenes, could be loaded into open pockets with high capacities. Notably, these hollow cage pockets provide multisites to capture different guests, showing heteroguest coloaded behavior to capture binary, ternary, or even quaternary guests. Moreover, a pair of commercially applied drugs for the combination therapy of chronic lymphocytic leukemia (CLL) has been tested, highlighting its potential in multidrug delivery for combined treatment.



INTRODUCTION

Inspired by the guest-binding pockets of enzymes and the hollow structures of protein cages like ferritins, viral capsids, clathrin, etc., synthetic chemists have ingeniously designed and assembled a variety of supramolecular cage receptors with varied sizes and shapes.^{1–6} The fruitful host–guest chemistry endows these artificial containers with unique capacities to bind a wide range of guest entities in enclosed cage cavities, including white phosphorus,^{7–9} polycyclic aromatic hydrocarbons,^{10–13} fullerenes,^{14–20} calixarenes,^{21–23} cryptophane,^{24–26} quantum dots,²⁷ and even proteins,²⁸ thereby opening up promising avenues for new applications.²⁹

Distinct from the conventional guest-binding behaviors of the internal cavities of most synthetic cages, many proteins possess multiple recognition sites in open solvent-accessible peripheral pockets, which are able to bind multiple guests or different types of guests, leading to diverse protein functions.^{30–32} Applying artificial receptors for high-order guest binding,^{29,33–40} especially multivariant or multicomponent guest binding, remains underdeveloped, although anomalous physicochemical properties or catalytic performance may be brought forth.^{18,41–47} On the one hand, synthesizing cages with multibinding sites requires elaborate structural design and preparation. On the other hand, reliable determination of high-guest-stoichiometry scenarios lacks effective analysis methods to acquire direct and valid binding

information, which typically relies on Nuclear Magnetic Resonance (NMR) or X-ray crystallography. Multiple-guest-included complexes often suffer from difficulty growing single crystals of suitable quality for X-ray diffraction. Besides, the solution characterization of either complicated protein–ligand interactions or dynamic host–guest interactions with multiple sites or guests is always challenging due to the fact that the limited and averaged spectroscopic information (NMR, UV–vis, etc.) makes it unexplicit to interpret nonspecific and differential multiguest-binding events, let alone fast guest exchange, broad line widths, and very short relaxation times often encountered in supramolecular systems,^{48–51} which sometimes are less informative and even cause misleading conclusions.^{49,50,52} Therefore, research into methodologies capable of directly providing binding information for high-order and heteroguest systems is of demanding interest. Mass spectrometry, with advantages of minimal sample consumption, rapid analysis, high sensitivity and resolution, and individual species identification, as well as well-developed

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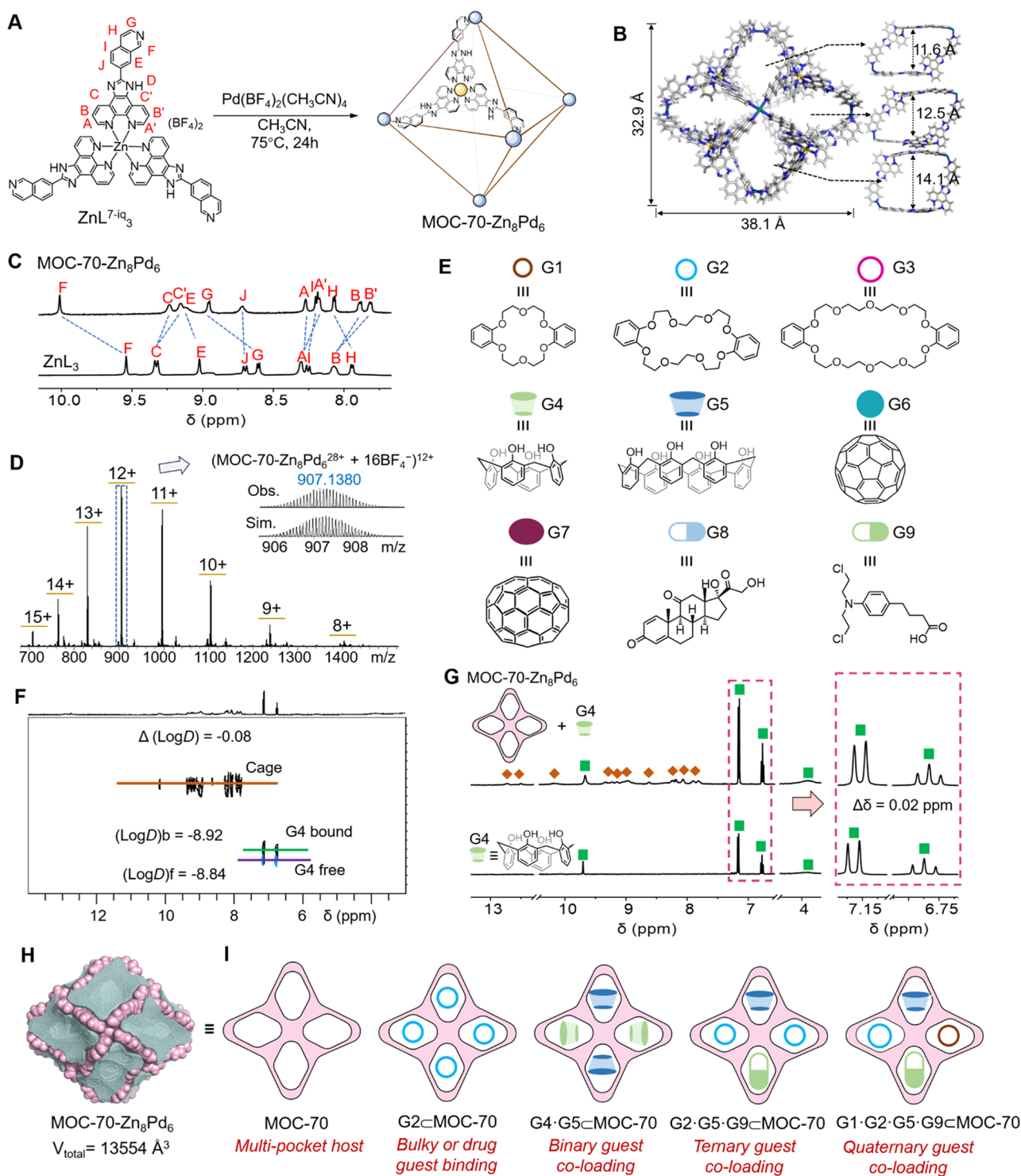


Figure 1. Synthesis, structure, and guest-binding modes of MOC-70-Zn₈Pd₆. (A) Synthetic route of MOC-70-Zn₈Pd₆. (B) Cage and pocket sizes of MOC-70-Zn₈Pd₆ based on a single-crystal structure. (C) Partial ¹H NMR of MOC-70-Zn₈Pd₆ and ZnL₃^{7-iq} metalloligand (CD₃CN/D₂O, 1/4, v/v). (D) HR-ESI-MS spectrum of MOC-70-Zn₈Pd₆. The inset shows the measured and simulated isotopic patterns. (E) Guest molecules. G1, dibenzo-18-crown-6 (brown circle); G2, dibenzo-24-crown-8 (blue circle); G3, dibenzo-30-crown-10 (magenta circle); G4, calix[4]arene (green cup); G5, calix[6]arene (blue cup); G6, C₆₀ (cyan solid circle); G7, C₇₀ (purple solid ellipse); G8, prednisone (blue capsule); G9, chlorambucil (green capsule). (F) Overlapped ¹H DOSY spectra of MOC-70-Zn₈Pd₆ + G4 and free G4 (diffusion coefficients of the cage, bound guest, and free guest are highlighted in color lines). (G) ¹H NMR spectra of free G4 and the G4-MOC-70-Zn₈Pd₆ complex. Signals of the guest are marked with solid green squares. The orange solid rhombus indicates the proton of MOC-70-Zn₈Pd₆. Proton peaks belonging to the bound guest are broadened and slightly upfield shifted compared to those of the free guest (CD₃CN, 400 MHz, 298 K). (H) Total cavity volume (pockets + inside cavity) of MOC-70-Zn₈Pd₆ calculated from the crystal structure. (I) Schematic diagram of the binding modes of bulky or drug guests, showing heteroguest coloads from binary to quaternary in the pockets of MOC-70-Zn₈Pd₆.

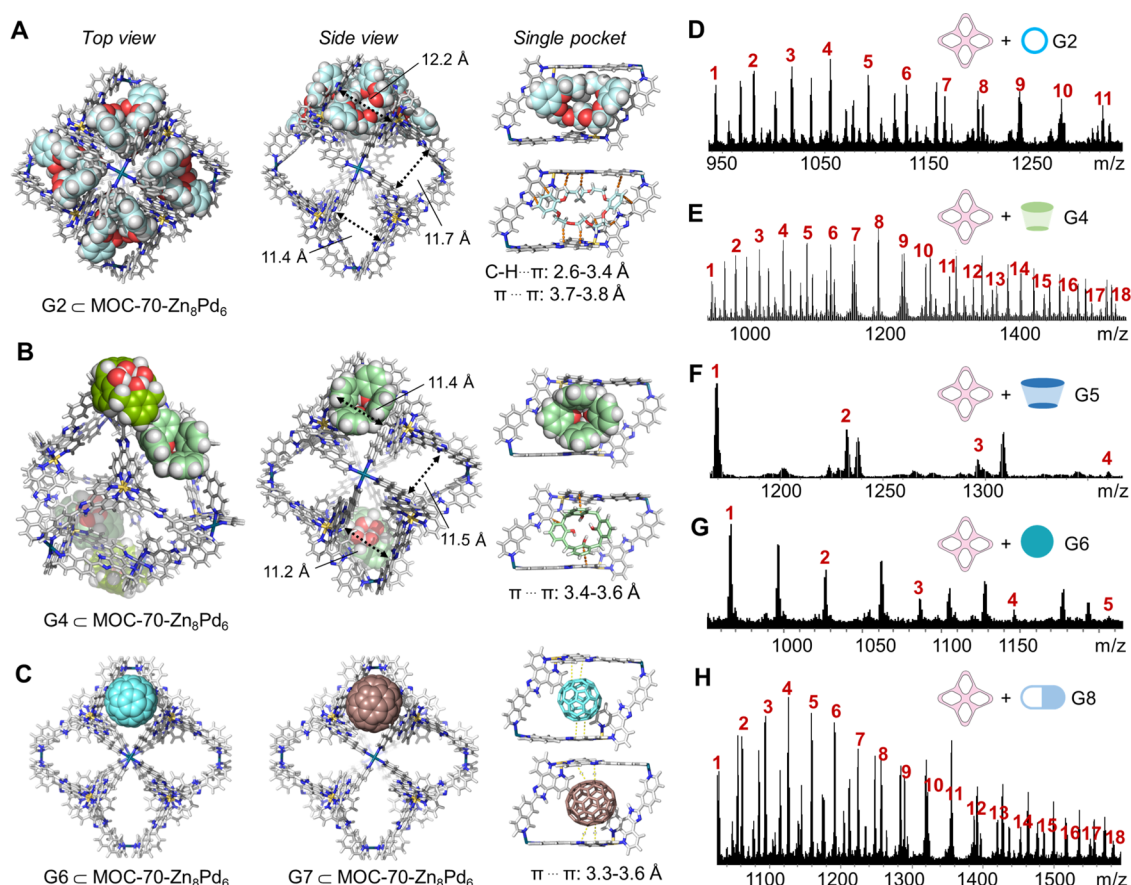


Figure 2. Unitary guest binding. (A, B) Single-crystal structures of G2 $\subset$ MOC-70-Zn₈Pd₆ and G4 $\subset$ MOC-70-Zn₈Pd₆ complexes. The size of the pockets, as well as the multiple π - π stacking and C-H $\cdots$ π interactions, is depicted. (C) Simulated structures based on the global energy minimization search and the extracted pockets of G6 $\subset$ MOC-70-Zn₈Pd₆ and G7 $\subset$ MOC-70-Zn₈Pd₆ complexes. (D-H) HR-ESI-MS spectra of representative G2 $\subset$ MOC-70-Zn₈Pd₆, G4 $\subset$ MOC-70-Zn₈Pd₆, G5 $\subset$ MOC-70-Zn₈Pd₆, G6 $\subset$ MOC-70-Zn₈Pd₆, and G8 $\subset$ MOC-70-Zn₈Pd₆ complexes (CH₃CN, 0.1 mM). Red numbers show the bound guests per cage.

techniques, provides another vivid characterization option, emerging as a valuable analytic tool not only suitable for studying high-stoichiometry guest-binding and distinct binding modes but also applicable to understanding the dynamic solution process free from solvent competition and heteroguest interference,^{12,45,53–60} albeit with the cautions of ionization and desolvation conditions in the gas phase.

Herein, we propose a mass spectrometric method to explore versatile multiguest-binding modes of a large-sized metal-organic cage (MOC-70-Zn₈Pd₆), which features 12 peripheral multipockets, directly observing high-capacity binding of drugs and bulky molecules (Figure 1), which is hardly accessible from other spectroscopic approaches. Notably, the dibenzo-crown ether and calixarene host molecules are found to fit in the spacious cage pockets to form host-in-host complexes. Coloading different types of guests leads to a wide repertoire of heteroguest-binding events, ranging from unitary to binary, ternary, and even potentially quaternary coencapsulation modes. Such diverse multiguest binding is challenging to achieve in the central cavity of a cage, which requires size, shape, and electronic nature complementary matching.^{61,62} Moreover, the dynamic multiguest uptake behaviors have been evaluated, suggesting distinct coinclusion processes depending on guest affinity. This work manifests a convenient yet powerful methodology to detect complex guest-binding scenarios, which is helpful in promoting new application studies of supramolecular receptors.

RESULTS AND DISCUSSION

Synthesis and Structure of MOC-70-Zn₈Pd₆. An elongated ZnL₃^{7-iq} metaloligand compared to our previously used ZnL₃³³ was prepared and assembled with Pd²⁺ in acetonitrile (CH₃CN) to result in formation of a huge supramolecular cage designated as MOC-70-Zn₈Pd₆ (Figures 1A and S1–S16 and Schemes S1 and S2). Clear ¹H NMR spectra indicated a convergent cage assembly in solution (Figure 1C), which was further confirmed by ¹H DOSY measurement to display a uniform size and shape with consistent diffusion coefficients of cage protons (Figure S11). A series of successively charged species attributable to [(ZnL₃^{7-iq})₈Pd₆ + *n*BF₄]^{(28–*n*)+} species were observed by high-resolution electrospray ionization mass spectrometry (HR-ESI-MS), indicating formation of a sole thermodynamically stable cage in CH₃CN that strongly withstands imidazole NH deprotonation in contrast to the smaller-sized MOC-16-Zn₈Pd₆ cage³³ (Figure 1D). However, when the solvent was changed from CH₃CN to the protic condition in a CH₃CN/CH₃OH = 2:1 (v/v) mixture, successively deprotonated species appeared due to imidazole–MeOH proton exchange (Figure S12). As a result, the proton-exchange process in solution and the dehydrogenation propensity under ionization conditions were largely inhibited in the aprotic solvent condition, which rendered much neat ¹H NMR resonance of MOC-70-Zn₈Pd₆ compared to MOC-16-Zn₈Pd₆ (Figure S13)

and significantly made MOC-70-Zn₈Pd₆ a good candidate cage to study host–guest complexation and guest-binding behaviors without severe interference of the mass peak analysis and interpretation arisen from dehydrogenation of the charged species to give complicated deprotonation states.^{63,64}

A rhombic dodecahedral cage structure of MOC-70-Zn₈Pd₆ was unambiguously established by single-crystal X-ray diffraction (Figures 1B, S14, and S15 and Table S1). Each cage is composed of 8 homochiral Δ_8 - or Λ_8 -ZnL₃^{7-iq} metaloligands and 6 Pd²⁺ ions, forming enantiomeric cage pairs of opposite chirality that cocrystallize in a single crystal via $\pi\cdots\pi$ stacking (Figure S14). A slight deformation from a regular rhombic dodecahedron in the solid state is observed for the cage dimensions of 38.1 × 32.9 × 32.9 Å measured from Pd⋯Pd distances, which leads to the inequivalence of the 12 pockets with the size and volume ranging from 11.6 to 14.1 Å and 734 to 905 Å³, respectively. The total volume of the 12 rhombic pockets and the central cavity of MOC-70-Zn₈Pd₆ is calculated to be 13554 Å³, nearly twice the size of analogous MOC-16-Zn₈Pd₆ (Figures 1H and S16).^{33,65} The remarkably enlarged open pockets endow MOC-70-Zn₈Pd₆ with great potential for accommodating bulky guest molecules.

Bulky Guest Loading by MOC-70-Zn₈Pd₆. Considering the fact that this cage features hollow cavity and multipockets with large sizes (narrowest width >1.1 nm), four types of bulky molecules were selected to test its guest inclusion capacity (Figure 1E): (1) three macrocyclic dibenzo-crown ethers, i.e., dibenzo-18-crown-6 (G1), dibenzo-24-crown-8 (G2), and dibenzo-30-crown-10 (G3), which are flexible host compounds typical for cation recognition; (2) two semiflexible aromatic calixarenes, i.e., calix[4]arene (G4) and calix[6]arene (G5), which are classic organic containers, (3) two rigid fullerenes (C₆₀, G6; C₇₀, G7), and (4) two drug molecules, i.e., prednisone (G8) and chlorambucil (G9), which are commercially available medicines proven effective in the combination therapy of chronic lymphocytic leukemia (CLL).⁶⁶ The host–guest interactions between these compounds and the MOC-70-Zn₈Pd₆ cage have been critically analyzed by means of NMR, HR-ESI-MS, X-ray crystallography, and simulations (Figures 2 and S17–S52), disclosing extraordinary high-order guest-loading capacity, unusual host-in-host accommodation, and, unprecedentedly, multiple-guest-loading ability of the MOC-70-Zn₈Pd₆ receptor.

Three dibenzo-crown ethers of different sizes showed similar binding behaviors toward MOC-70-Zn₈Pd₆, as its rhombic pockets are long enough to accommodate these flexible macrocycles without significant conformation adjustment. The proton resonances of the bound guests became broad and slightly shifted upfield in comparison with those of free guests due to the aromatic shielding effect by the pocket walls (Figures S17–S19). The host–guest interactions were also testified by ¹H DOSY measurements, where the diffusion coefficients (Log *D*) of the bound guests fall between the cage and the free dibenzo-crown ethers (Figures S20–S22). The appearance of one set of ¹H NMR signals of both cage and guests suggested a rapid guest-exchange process in solution on the NMR time scale, which was further confirmed by the differentiated Log *D* values observed for the bound and free guests.^{51,67} The titration experiments of these guests into a cage solution indicated that the upfield shifts of the guest signals gradually tapered off along continual additions. Although the resonance variation ($\Delta\delta$) generally exhibited break points to suggest dynamic multiguest-binding equilibria

in solution,³³ the large open pockets embedded on the cage sphere are expected to render a faster guest-exchange rate and a weaker binding affinity of bulky or flexible guests than the inner cage cavity, resulting in small $\Delta\delta$ changes that make data fitting inexplicit by using the known analysis models, such as Hill equation, Job plot, and BindFit, which are more suitable for the low guest binding with significant affinity,^{49–52} to obtain reliable association constants and stoichiometry (Figures S23–S25). Moreover, the multiguest and multi-component binding events are hard to differentiate by the overall averaged and broadened proton resonances; therefore, acquiring informative binding isotherms for such rapidly exchanging and multiple-guest-binding supramolecular systems by virtue of the NMR technique is apparently challenging.

To our delight, the single crystals of the G2-bound G2 ⊂ MOC-70-Zn₈Pd₆ (G2 ⊂ H, similar hereafter) complex were obtained, and the X-ray diffraction analysis revealed the binding mode between dibenzo-crown ether and the cage (Figures 2A and S26 and Table S2). When capturing G2 guests, the cage pockets adjust a little bit to a more uniform conformation with only four top pockets slightly expanded to accommodate four G2 molecules separately. The $\pi\cdots\pi$ stacking between the benzene rings of G2 and the two trapezoidal-shaped ligand walls of the rhomboid pocket of the cage drives the binding of the guests. The motion of the oxyalkyl chain is restricted through multiple C–H⋯ π interactions with two parallel pocket walls. These observations suggest that the cage may turn more symmetric in solution with all pockets readily adaptive to bulky guest binding via a dynamic fit process.³³ Occupation of four G2 guests in the four top pockets only slightly compresses other empty ones, which may somewhat retard following guest binding but should not significantly preclude the cage from capturing more guests in solution, due to the well-known induced-fit effect of the cage host.²² This structural analysis forecasts a plausible high-order guest-binding scenario in solution, as observed from an HR-ESI-MS investigation.

Taking advantage of high sensitivity toward the isolated structural species despite their distribution under dynamic equilibrium and fast interguest exchange, HR-ESI-MS is capable of precisely detecting various guest-loaded complexes, thereof commonly employed in the pharmaceutical study of the solution host–guest chemistry⁵⁵ to provide valuable binding evidence and determine the stoichiometry of protein–ligand interactions. As seen from Figures 2D and S27–S29, similar guest-binding profiles were observed for MOC-70-Zn₈Pd₆ in the presence of three differently sized flexible crown ethers. A high-stoichiometry guest inclusion up to a guest-to-host ratio (G/H) of 12:1 was observed, showing a successive distribution of the guest-loaded complexes with maximal mass peaks centered around G/H = 2–4. Considering possible partial guest dissociation during ionization, these guest-binding profiles may reflect the host–guest interacting process in solution to a close reality, consistent with the guest titration results that the proton $\Delta\delta$ shifts showed breaking points in the G/H = 2–6 range (Figures S23–S25), indicating dynamic multiguest-binding equilibria in solution centered around these host–guest stoichiometries.³³ As learned from the crystal structure of the G2 ⊂ H complex, the 12 cage pockets may initially uptake 4 guest molecules to slightly deform cage conformation, slowing down the guest binding of the remaining 8 pockets. However, capturing 8 guest molecules into these 8 pockets still occurs via mutual dynamic fit with

A Binary quest co-loading

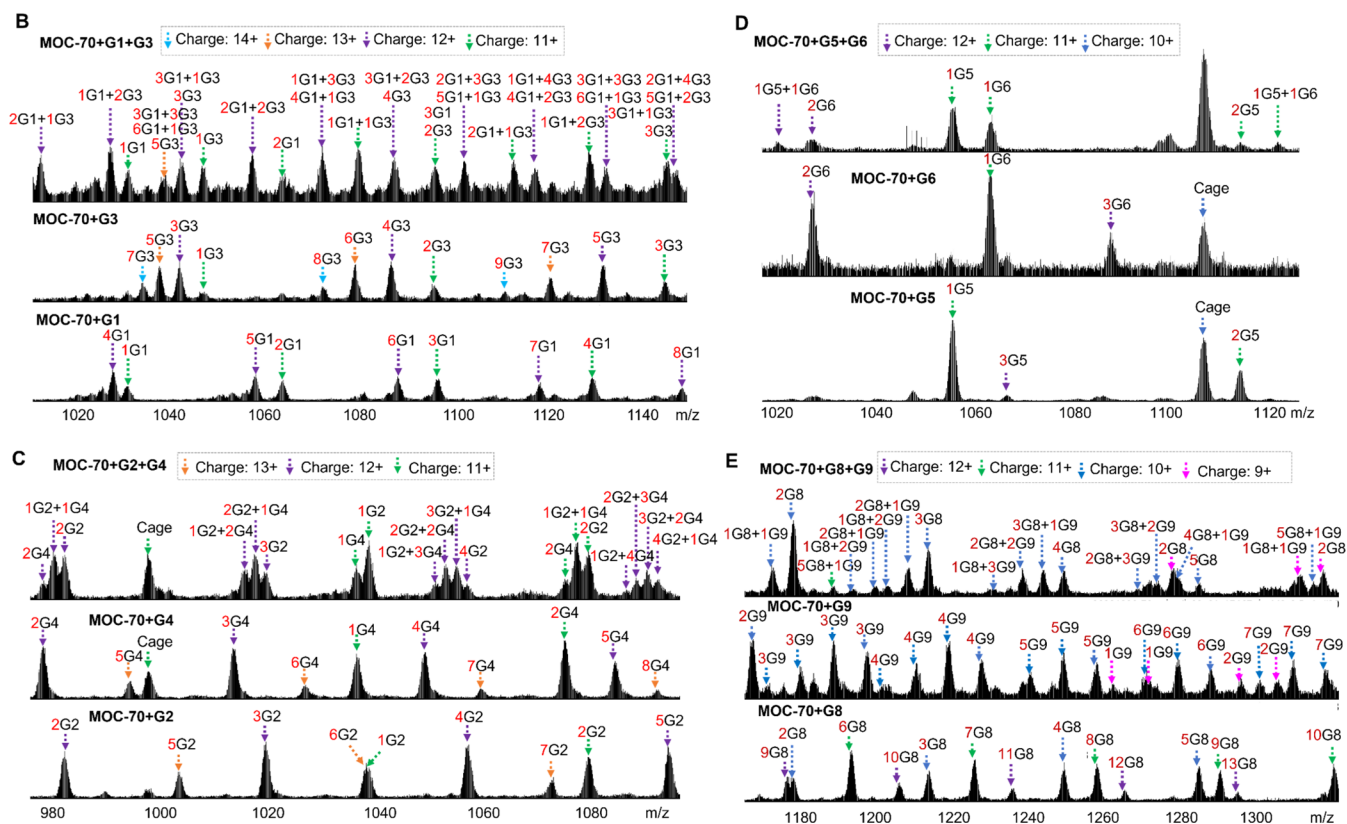
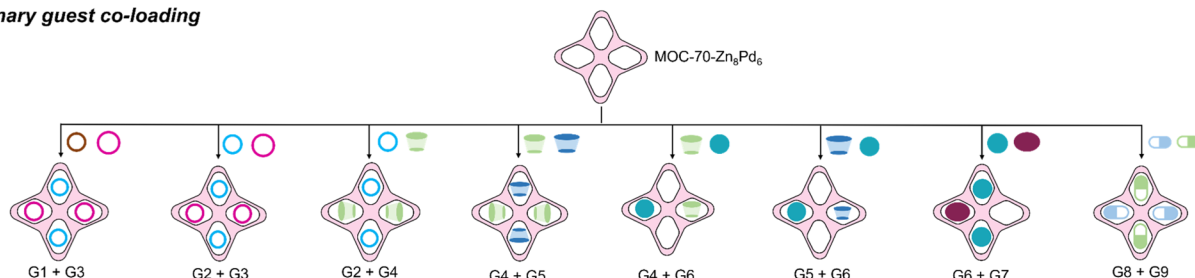


Figure 3. Binary guest coloaded determined by HR-ESI-MS analyses. (A) Schematic diagrams showing binary guest coloaded modes. (B–E) Partial HR-ESI-MS spectra comparisons between the unitary guest loading in MOC-70-Zn₈Pd₆ and the binary guest coloaded in MOC-70-Zn₈Pd₆. G_{mixed}/H = 15:1. The ratio between guests is 1:1.

probably a lower affinity. The similarity in binding profiles may stem from the flexible nature of the three guests, enabling adaptability to the pocket shape and space.

The semirigid aromatic guest G4, which was found to act as a mutual induced-fit epitome in host–guest interactions,^{22,33} was further applied for MOC-70-Zn₈Pd₆ inclusion (Figures 1F,G, 2B,E, and S30–S33). The NMR studies suggested fast interguest exchange on the NMR time scale. However, single-crystal X-ray diffraction analysis of the G4CH complex indicated that G4 perfectly fit into the cage pockets without deformation of the cage (Figures 2B and S33 and Table S3). Therefore, MOC-70-Zn₈Pd₆ demonstrates excellent G4-binding ability, showing a wide range of up to G/H = 18:1 guest-binding species in solution (Figures 2E and S34). Such an excessive guest-binding ratio compared to bulky macrocycles G1–G3 originates from the peripheral binding of G4 guests except those captured in 12 pockets. As observed in the single-crystal structures (Figures 2B and S33), G4 can either be encapsulated in the 12 cage pockets or cap the 6 vertices from the outside of the cage, thereof forming both host-in-host and

host-on-host binding modes. In contrast, a pronounced size-dependent binding behavior was found for the larger G5 calixarene (Figures 2F and S35–S38), displaying a much less binding capacity of G/H = 4:1 in the MS spectral profile due to the poorer binding affinity compared to G4.

The rigid spherical fullerene guests were also tested for encapsulation by MOC-70-Zn₈Pd₆ (Figures 2C,G and S39–S44). The HR-ESI–MS and ¹³C NMR revealed that C₆₀ and C₇₀ could be loaded into the cage; however, a low G/H ratio of 5:1 was observed for C₆₀ based on MS measurement, while that of C₇₀ decreased to 2:1 (Figures 2G, S39, and S40), resembling fullerene binding of recently reported cages with multiple sites.⁴⁰ A global energy minimization search was performed to locate the fullerene position (Figures 2C and S43), which suggested that the cage is adaptable to C₆₀ or C₇₀ at both the inner cavity and open pockets. Nevertheless, the pocket sizes would be slightly expanded via $\pi\cdots\pi$ stacking to fit with C₆₀ and C₇₀,¹⁴ which may account for the low loading capacity of the MOC-70-Zn₈Pd₆ cage for these two rigid spherical guests' lack of adaptiveness. The solution-state host–

guest interactions of the MOC-70-Zn₈Pd₆ cage with the G6 guest were further testified through luminescence and UV–vis absorption titration experiments (Figure S44), indicating that each cage binds approximately 4–5 guests, which is consistent with the mass spectrometric results.

Finally, the two commercially available medicines G8 and G9 were subject to guest-binding evaluation (Figures 2H and S45–S52). Although fast exchange binding behaviors were found for G8 and G9 guests, they were capable of being loaded into a MOC-70-Zn₈Pd₆ cage with high-order capacity. HR-ESI–MS investigations revealed similar G8 and G9 capturing profiles with those of G1–G4 guests, demonstrating excellent loading capacity, reaching G/H ratios as high as 18:1 and 16:1 for G8 and G9, respectively (Figures 2H, S51, and S52), despite the immediate weakening of MS peaks with the increase of the bound guests for G9.

The above HR-ESI–MS results indicate that this multi-pocket cage exhibits a high binding ratio for guests, depending on their different sizes, shapes, and flexibility. Compared to guest encapsulation inside the central cavity of the cage, higher guest-binding capacity and complexity could be achieved; however, a relatively weaker binding affinity and a faster guest-exchange rate are also expected for open peripheral pockets, which facilitate dynamic equilibrium between the bound and free guests in solution to establish a statistic distribution of guest-binding stoichiometries. These variant guest-binding events are hard to acquire by traditional ¹H NMR titration or DOSY measurements, since the entirely averaged and poorly resolved proton resonances are not adequate for providing direct stoichiometric information and distinguishing varied binding events between multiple sites and guests. In this context, the mass spectrometric method is unique and valuable for the understanding and determination of high stoichiometric multiguest and multisite binding modes in solution.

Binary Guest Coloading of MOC-70-Zn₈Pd₆. Given this hollow cage possesses an array of open pockets adaptive to diverse bulky guests with a high-order capacity, we are interested if it can act as a supramolecular carrier to load heteroguests (Figure 11). We first tried binary coloading composites with various combinations of the aforementioned molecules. Thanks to the powerful ability of the HR-ESI–MS technique to identify individual molecular ensembles simultaneously and quickly, we detected a full spectrum of distributions of dual guest loading signals by directly mixing MOC-70-Zn₈Pd₆ with two different guests (Figures 3 and S53–S76), which is inaccessible by other characterization methods in terms of coexisting multivariant or multi-component guest-binding events.

For guests G1, G2, G3, and G4, which are capable of achieving high stoichiometric binding individually in MOC-70-Zn₈Pd₆, a series of binary coloading species with variable compositions were obtained when adding the G1/G3, G2/G3, or G2/G4 mixture into the MOC-70-Zn₈Pd₆ solution (Figures 3B,C and S54–S61). Although a slight excess of mixed guests at a 1:1 ratio was added (15 equiv of G_{mixed} per cage vs 12 pockets), the fully loaded host–guest complexes were absent, probably due to ionization conditions in the gas phase. For instance, the highest compositions of 5G1 + 2G3, 2G2 + 3G4, or 5G2 + 3G3 were observed for G1/G3, G2/G4, or G2/G3 coloading, respectively, while abundant low-loading combinations appeared as commonly encountered in MS conditions. Moreover, most of the unitary guest-loading species disappeared in comparison with the separated single-guest plus

cage systems. These results suggest that MOC-70-Zn₈Pd₆ not only is efficient in individually multiple guests binding but also has a superior heteroguest coloading capacity, undergoing either heteromolecular guest exchange or sequential filling into the unoccupied pockets in solution when adding mixed guests. It is worth noting that there are some limitations of this mass spectrometric method, especially for distinguishing different entities with overlapping mass-to-charge ratios. For example, in the case of G1/G3, the approximate equality of M_w (3·G1) and M_w (2·G3), as well as the broad overlap in their isotopic distributions, makes it challenging to assign exactly just based on the mass spectrometry. Therefore, possible compositions of different binary guest-loading complexes are tentatively designated for some MS peaks.

Calixarenes G4 and G5 exhibit significantly distinct unitary binding ability; however, coloaded of these two container molecules into MOC-70-Zn₈Pd₆ pockets was evidently confirmed by the MS measurement, despite the fact that many peaks could not be precisely discriminated due to overlapped guest combinations with almost identical molecular weights (Figure S64). Noteworthy, the coloading process apparently favors the right sized G4 compared to bigger G5, resulting in relatively high G4 loading with two guests summing up to 11. On the contrary, rigid fullerenes G6 and G7 only gave low-coloading complexes, such as (1G6 + 1G7) ⊂ H and (2G6 + 1G7) ⊂ H (Figure S67). Moreover, the prominent unitary guest-binding peaks corresponding to the single-guest plus cage systems were still present, suggesting the poor coloading ability of these two undeformable guests. Similar coloading behavior was even observed for G4/G6 and G5/G6 combinations, giving just simple (1G4 + 1G6) ⊂ H and (1G5 + 1G6) ⊂ H complexes (Figures 3D, S70, and S73). This means the coloading capacity of MOC-70-Zn₈Pd₆ is subject to mutual size matching and adaptiveness between cage pockets and guests.

The combination therapy of different medicines represents a promising approach for cancer treatment, imparting enhanced efficacy, decreased toxicity, and reduced drug resistance.⁶⁸ Loading the drugs for combination therapy onto the same carrier can ensure that both drugs are delivered to the same cells in specific ratios and have consistent pharmacokinetic profiles, which may offer advantages for combination therapy.^{69,70} Currently, most drug codelivery systems are based on liposomes, micelles, and nanoparticles. However, codelivery using single-molecule platforms like metal–organic cages has rarely been reported. As MOC-70-Zn₈Pd₆ displayed excellent multiple-guest-binding capability, we tested drug coloading with a pair of commercial medicines G8/G9 for the potential of MOC-70-Zn₈Pd₆ as a cotransporting cargo (Figures 3E and S74–S76).⁶⁶ Despite the size disparity, G8 and G9 showed notable coloading capability when equal amounts were mixed into the MOC-70-Zn₈Pd₆ solution, as demonstrated by a series of heteroguest-bound MS peaks (Figures 3E, S75, and 76). It was noticeable that the G8-bound complex was observed under the G8/G9 = 1:1 condition, and most MS peaks were assignable to compositions with more G8-bound guests than G9. This suggests a stronger host–guest affinity for G8 than for G9, probably because G9 is conformationally too flexible and has fewer hydrogen-binding groups to interact with imidazole N/NH sites than G8, thus undergoing faster interguest exchange in solution.

The unique feature of MOC-70-Zn₈Pd₆ for heteroguest coloading is that it possesses multiple open pockets on the cage

A High-order hetero-guest co-loading

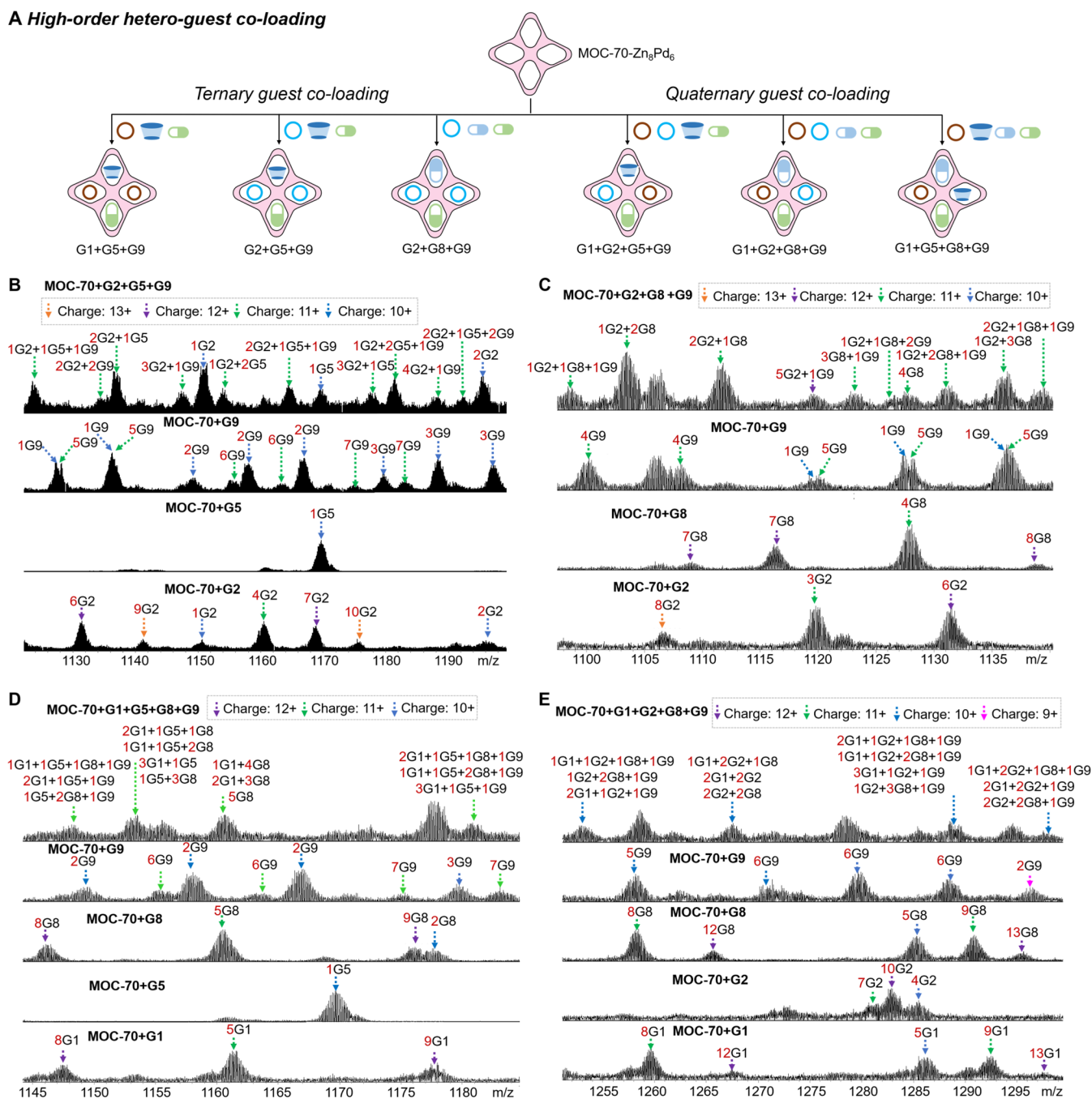


Figure 4. High-order heteroguest coloaded determined by HR-ESI-MS analyses. (A) Schematic diagram showing the high-order heteroguest coloaded modes. (B–E) Partial HR-ESI-MS spectral comparisons of the unitary guest loading in MOC-70-Zn₈Pd₆ and the ternary or quaternary guest loading in MOC-70-Zn₈Pd₆. $G_{\text{mixed}}/H = 15:1$. The ratio between guests is 1:1:1 or 1:1:1:1 for ternary and quaternary heteroguests, respectively.

sphere. Although the central-cavity accommodation of the guests should not be excluded, the high stoichiometric guest-binding capacity is most likely contributed by the peripheral pockets as previously demonstrated by Reek, Nitschke, and us.^{33,40,45,65} Different from the normal heteroguest recognition model in the inner cage cavity,²⁹ which necessitates adaptive and complementary matching not only between the host and guest but also among the guests themselves,^{62,71,72} the multiple pockets on this cage work separately in an overall synergistic action. The spacious pocket cavities are able to capture bulky guest molecules of varied sizes and shapes, while the interactive pocket conformations render the ability to bind different guests

with a wide range of combinations and variations. Therefore, this type of cage allows for the coloaded of heteroguests with less limitation on the mutual matching consideration between the guests, facilitating a high loading ratio and diverse guest selection.

High-Order Heteroguest Coloaded of MOC-70-Zn₈Pd₆. The detection of even higher stoichiometric guest-binding modes becomes more challenging due to the weakened MS peaks and severe overlapping resulting from the increased diversity of bound species (Figures 4 and S77–S94). Nonetheless, we successfully achieved the characterization of the ternary guest coloaded by the MOC-70-Zn₈Pd₆

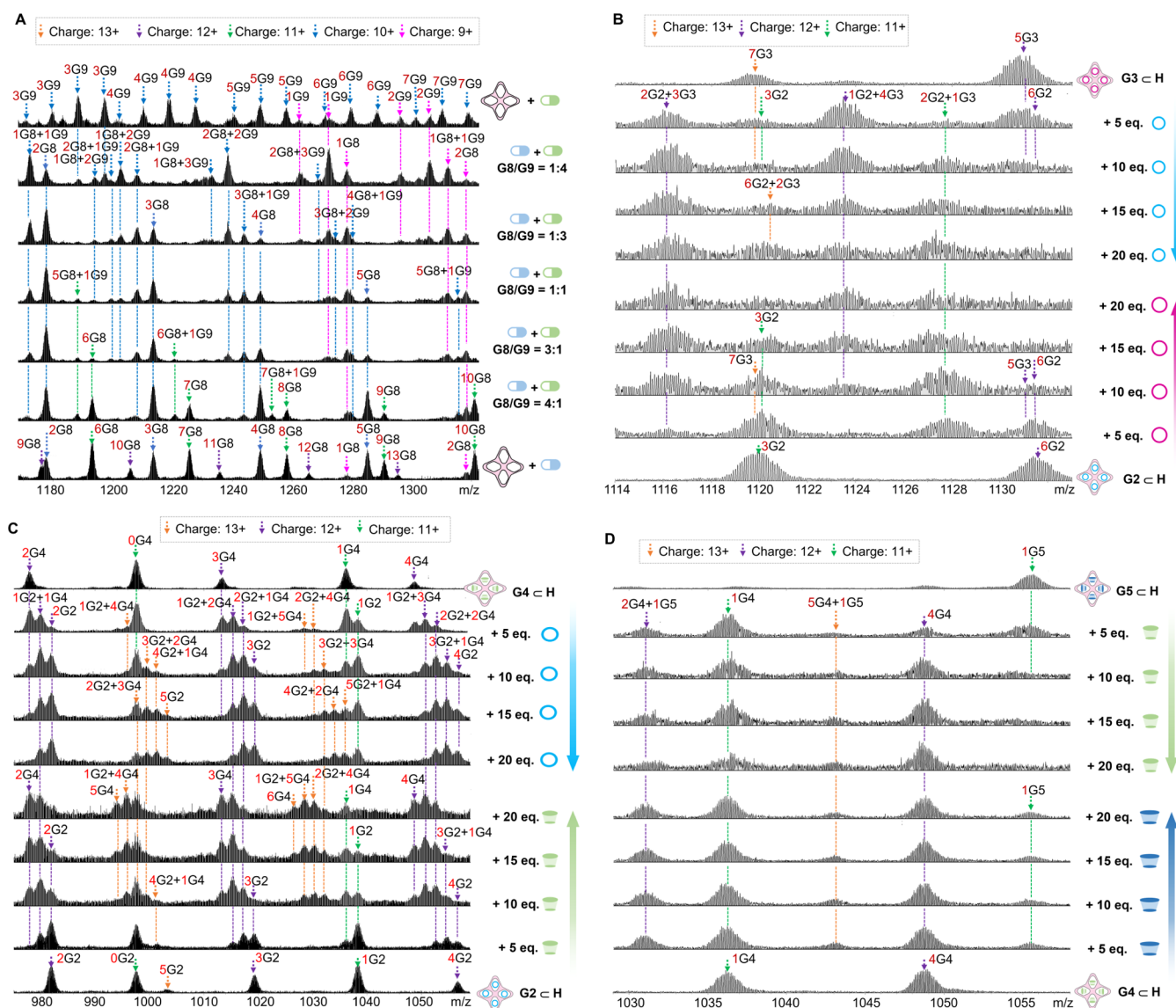


Figure 5. Heteroguest coload dynamics. (A) Comparison of partial HR-ESI-MS spectra of the MOC-70-Zn₈Pd₆+G8+G9 system with varied G8/G9 ratios. $G_{\text{mixed}}/H = 15:1$. (B–D) Partial HR-ESI-MS titration spectra of single-guest-loaded systems ($G/H = 15:1$) with secondary guest addition in both forward and backward fashions. The numbers in red show the guest-binding ratio per cage.

cage with the HR-ESI-MS method. The clearly identified MS peaks of G1/G5/G9, G2/G5/G9, and G2/G8/G9 (1:1:1, 15 equiv. G_{mixed} per cage vs 12 pockets) guest combinations were demonstrated in Figures 4B,C and S77–S85. Regardless of the dual assignment of a few MS peaks, most of the salient signals could be exactly correlated with binary and ternary guest-binding events, as confirmed by a comparison between the measured and simulated isotopic patterns. General lower guest numbers were observed as compared with the binary coload systems, suggesting a propensity of guest dissociation under ionization conditions for the more complex ternary host–guest systems. The survival of even complex multiguest-binding species under MS conditions, in the case of quaternary guest coload, seems to reach the limit of the mass spectrometric method. The introduction of one additional component into the ternary systems resulted in completely overlapped signals (Figures 4D,E and S86–S94), which made reliable discernability difficult due to closely resembling molecular weights of different guest combinations, e.g., $M_w (1G1 + 1G5 + 2G8 +$

$1G9) \approx M_w (3G1 + 1G5 + 1G9)$ for G1/G5/G8/G9 and $M_w (1G1 + 1G2 + 1G5 + 1G9) \approx M_w (4G1 + 1G9)$ for G1/G2/G5/G9 systems. This MS limitation prevented us from convincing determination of the coload capability of the MOC-70-Zn₈Pd₆ cage for more than three different guests because the 4-guest-binding complexes could not be unambiguously discriminated from the 2- or 3-guest-binding possibilities. Nevertheless, the distinct MS profiles between the ternary and quaternary systems containing similar guest components imply that MOC-70-Zn₈Pd₆ still holds the potential to bind more diverse guests. To our knowledge, only a few cage examples were reported to be capable of capturing different guests,^{12,18,25,43,72–77} and the high-order heteroguest loading of big molecules remains scarce, signifying the large multipocket cage model like MOC-70-Zn₈Pd₆ as promising carriers of medicine guests with variable sizes and shapes.

Heteroguests Uptake Process in Solution. To further understand the coload dynamic behaviors of the host cage

toward different guest molecules, more detailed MS exploration was performed for binary systems with varied ratios of heteroguests (Figures 5A and S95, S96) or with heteromolecular titrations in both forward and backward ways by adding the secondary guest (0–20 equiv) to a solution containing the first guest (15 equiv, Figures 5B–D and S97–S108). For a binary host–guest system showing comparable guest-binding properties like $G2/G3 \subset H$, continuously adding the secondary guest up to 20 equiv to the single-guest solution led to the disappearance of the first-guest-bound species quickly (Figures 5B and S97–S100), irrespective of whether starting from $G2 \subset H$ or $G3 \subset H$ in the presence of 15 equiv of the first guest. Concurrently, the emergence of a series of heteromolecular coloaded peaks was observed. This phenomenon is attributed to the sequential binding of the secondary guest with fast guest exchange that leaves empty pockets or replacement binding driven by heteromolecular guest exchange due to comparable binding ability, suggesting a strong propensity to coload different guests by the cage to establish a heteroguest-binding equilibrium.

For a binary $G2/G4 \subset H$ system with distinct guest shape and rigidity, we observed the gradual vanishing of the $G4 \subset H$ signals and emerging of $G2/G4$ cobinding species, as well as the appearance of new signals belonging to $G2 \subset H$, with addition of excessive $G2$, and vice versa (Figures 5C and S101–S104). Unlike the $G2/G3 \subset H$ mixture, where rapid exchange between heteroguests leads to the quick disappearance of initial unitary signals, the presence of $G2 \subset H$ or $G4 \subset H$ unitary species with addition of the secondary $G4$ or $G2$ guest, respectively, indicates that the heteromolecular exchange between these two structurally distinct guests is relatively slower than that between structurally resembling $G2$ and $G3$ guests in solution, which may permit the preservation of all transient states under MS conditions to be semiquantitatively evaluated to some extent.^{60,78} Therefore, the nearly equal binding affinities of two guests render a statistical distribution of $G2/G4 \subset H$ complexes related to the $G2/G4$ ratios.

We then tested a pair of guests $G4/G5$ with significantly different binding capacities (Figures 5D and S105–S108). Given the less matched characteristic of larger guest $G5$, addition of $G5$ into the $G4 \subset H$ solution did not noticeably affect the $G4 \subset H$ signals; meanwhile, the new signals corresponding to $G4/G5 \subset H$ coloaded complexes appeared and changed little with $G5$ addition (Figures 5D and S105, S106). This suggests that the $G5$ guest has a poor ability to exchange $G4$ from the pockets, and heteroguest coloaded quickly achieves equilibrium with only a small equivalence of $G5$ uptake. On the contrary, when $G4$ was added to the $G5 \subset H$ solution, the signals of $G5 \subset H$ decreased rapidly and completely disappeared after addition of 10 equiv of the $G4$ guest. Instead, the $G4 \subset H$ species and $G4/G5 \subset H$ coloaded complexes emerged (Figures 5D and S107, S108), implying a fast shift of the guest-binding equilibrium toward $G4$ uptake.

The dynamic coloaded processes provide a chance to regulate the heteroguest ratios within the cage host to meet varying dosage requirements of drugs. We found that the coloaded capacity of $G8/G9$ could be tuned readily by adjusting the drug-adding proportions of $G8/G9$ (4:1–1:4, Figures 5A, S95, and S96). In the case of a significant excess of $G9$ ($G8/G9 = 1:4$), several $G8 \subset H$ species were still present. In contrast, an excess of $G8$ ($G8/G9 = 4:1$) caused the almost vanishing of $G9 \subset H$ species. This is consistent with the binary guest coloaded experimental result that $G8$ was favorably

loaded. Nevertheless, with alteration of the $G8/G9$ ratios, continuous regulation of the overall coloaded distribution of these two drugs was achieved, as seen from the gradual variations of the heteroguest coloaded MS peaks. This highlights the unique advantage of the mass spectrometric method in analyzing the binding behaviors of individual drug molecules for the assessment of their coloaded capability.

CONCLUSIONS

We established a multipocket supramolecular cage model for high-capacity accommodation of a variety of bulky and drug molecules with distinct sizes, shapes, and flexibility and demonstrated unique application of the HR-ESI–MS method in detecting multiple-guest-binding scenario and high-order coloaded of heteromolecular guests by a single cage. The multibinding sites in large hollow MOC-70- Zn_8Pd_6 facilitate capture and guest exchange in its 12 peripheral open pockets, which work individually in a synergistically concerted fashion to enable heteroguest coloaded through binary, ternary, and even quaternary component uptake, avoiding the strict restriction on guest–guest complementary needed for the inner cavity encapsulation of multiple guests. Noteworthy, the HR-ESI–MS technique is proven effective in characterization of complex guest-binding behaviors of supramolecular systems, achieving sensitive identification of the multiguest interactions with the host and discrimination of heteroguest-binding events within the same cage, which has been unattainable by other spectroscopic methods so far. This work provides a promising direction to design easy-to-make cage structures for complicated multiguest binding and heteroguest coloaded tasks and sheds light on studying problematic host–guest interactions by utilizing the MS approach to afford more informative and precise analysis, which is helpful in pushing the research of cage chemistry toward new applications such as multidrug codelivery for combination therapy.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c05758>.

Experimental procedures; crystallographic data; and additional host–guest NMR and HR-ESI–MS spectra. (PDF)

Accession Codes

CCDC 2324469–2324471 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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